

A380X computed alternate fuel and EXTRA time

The ALTN fuel and time of FUEL&LOAD computed as the FCOM defines them, instead of a fixed 6.5 t; the EXTRA time next to the EXTRA fuel

Aircraft	A380X
Where to find it	MFD > ACTIVE > FUEL&LOAD (ALTN line) and ACTIVE > F-PLN > ALTERNATE
PR branch	pr/20-a380x-altn-fuel-computation (stacked on PR 10)
Status	Built with 5 unit tests; ALTN checked in the sim during the fuel formula audit (September 2026); EXTRA time checked in the harness only.
Pictures	Captain MFD in the local browser harness (no flight plan, so the values are dashes). The table in section 3 comes from the predictor of this feature run on its own.

1. What it is for

2. Where you see it

3. How much to expect

4. The computation
5. EXTRA time

6. What to expect

7. Limitations and why

1. What it is for

The block fuel includes the fuel to divert to the alternate. FBW used a fixed 6.5 t, whatever the alternate. The A380 FCOM (DSC-22-FMS-20-30, FUEL&LOAD page) defines the ALTN fuel as the fuel (and time) for the trip from the primary destination to the landing at the alternate, computed with **CI 0**, at **FL220** when the alternate route is shorter than 200 NM and **FL310** otherwise. This is what the FMS computes now.

2. Where you see it

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/FUEL&LOAD

GW

CG

FOB

ZFW

ZFWCG

BLOCK

FUEL PLANNING *

TAXI

PAX NBR

TRIP

CI

RTE RSV

JTSN GW

ALTN

TOW

FINAL

LW

UTC

EFOB

MIN FUEL AT DEST

DEST

ALTN

EXTRA

RETURN

MSG LIST

Figure 1. FUEL&LOAD: the ALTN line gets its fuel and its time; the arrival at the alternate (destination + ALTN time) is under ALTN in the UTC column.

- The distance is the alternate flight plan distance, without its missed approach; without an alternate route, the direct distance.
- The weight is the weight at the destination (ZFW + MIN FUEL AT DEST), with the ALTN wind of the WIND page.
- No ZFW: no default ALTN fuel. The value is recomputed only when its inputs change.

3. How much to expect

Results of the predictor for an alternate close to sea level, fuel on board 12.5 t at the destination (fuel in tonnes / time in minutes / cruise level):

Alternate distance	ZFW 300 t	ZFW 344 t	ZFW 344 t, 30 kt headwind	ZFW 344 t, 30 kt tailwind
60 NM	2.3 t / 13 min / FL140	2.3 / 13 / FL120	2.5 / 14	2.1 / 12
100 NM	3.7 / 19 / FL220	3.7 / 19 / FL200	4.0 / 21	3.4 / 18
150 NM	4.9 / 26 / FL220	5.1 / 27 / FL220	5.5 / 29	4.8 / 25
199 NM	6.1 / 34 / FL220	6.3 / 34 / FL220	6.8 / 37	5.9 / 32
200 NM	6.3 / 32 / FL310	6.5 / 33 / FL310	6.9 / 36	6.2 / 31
300 NM	8.5 / 46 / FL310	8.8 / 46 / FL310	9.4 / 50	8.3 / 44
400 NM	10.8 / 59 / FL310	11.1 / 59 / FL310	11.8 / 64	10.4 / 56

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/F-PLN/ALTERNATE

DATA FROM

TO ALTN

CO RTE

DIR TRK

DIST

EXTRA

SELECTED

DATABASE

NO ALTN

OTHER ALTN

RETURN

MSG LIST

Figure 2. ALTERNATE page: the fuel to each alternate (EXTRA column) uses the same computation.

500 NM	13.0 / 72 / FL310	13.3 / 73 / FL310	14.2 / 78	12.5 / 68
--------	-------------------	-------------------	-----------	-----------

- At 200 NM the level jumps from FL220 to FL310 (the FCOM rule): the time drops a little, the fuel rises a little.
- Short alternates (60-100 NM) fly lower than FL220: the cruise level is lowered until the climb and the descent fit in the distance.
- The old fixed value, 6.5 t, is about right only for an alternate 200 NM away.

4. The computation

- Climb at climb thrust, cruise, idle descent, with the aircraft's VNAV prediction models (the same as the FMS predictions).
- Climb speeds of the A380 FMS: 290 kt climb and cruise, 288 kt descent, M0.84, 250 kt below FL100.

5. EXTRA time

The EXTRA time next to the EXTRA fuel (FCOM: the fuel, and time, that remains after landing at the destination, minus the MIN FUEL AT DEST) was never computed and always showed --:--. It is now the time the EXTRA fuel lasts at the holding fuel flow of the FINAL fuel default, **0.2 t/min** (30 min = 6 t): 7.4 t of EXTRA shows 00:37, 12.0 t shows 01:00.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/FUEL&LOAD

GW

CG

FOB

ZFW

ZFWCG

BLOCK

FUEL PLANNING *

TAXI

PAX NBR

TRIP

CI

RTE RSV

JTSN GW

ALTN

TOW

FINAL

LW

TIME

EFOB

MIN FUEL AT DEST

DEST

ALTN

EXTRA

RETURN

MSG LIST

Figure 3. FUEL&LOAD with 7.4 t of EXTRA fuel: EXTRA time 00:37. The harness has no flight plan, so the EXTRA fuel is set by hand for this picture; the other values stay as dashes.

- The FCOM leaves the FINAL default to the airline fuel policy (AMI file) and gives no rule to turn the EXTRA fuel into a time: the same holding flow keeps FINAL and EXTRA consistent on the page.

- No EXTRA fuel (no flight plan predictions, or a SEC flight plan): --:-- , as the fuel.

6. What to expect

Situation	What you should see
No ZFW entered	ALTN fuel shown as dashes (no default).
Alternate route entered	ALTN fuel and time from its distance; the arrival time at the alternate in UTC.
No alternate route, alternate airport set	Computed on the direct distance.
ALTN wind entered	More fuel with a headwind, less with a tailwind.
EXTRA fuel shown	EXTRA time = EXTRA fuel at 0.2 t/min (6 t = 30 min).

7. Limitations and why

Limitation	Why	What to do
One mean wind for the alternate trip.	The FCOM computation uses the ALTN wind of the WIND page.	-
No holding or approach fuel in ALTN.	The FCOM definition ends at the landing at the alternate; FINAL covers the reserve.	-
EXTRA time at a fixed holding flow.	The FCOM gives no fuel-to-time rule for EXTRA; the flow is the FINAL default one, not a computed holding at the actual weight.	-